

Vivekananda Library

MAD-1 Library Management System Project Report

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Description:

In this library, both librarian and general user's login to their respective dashboards. The librarian can access all the sections and books in each section and can perform edit, delete, create operations on them. He can also track the details of user and book requests from user. User can request, return, give feedback and access books, given access from the librarian.

Frameworks used:

Flask – used to build the application

Flask-SQLAlchemy – to interact with database

SQLite – database used for data storage

Bootstrap – used for HTML generation and styling

Jinja2 – python library used for templating

Architecture:

“app.py” – contains main flask application.

“templates” – all the html files reside in this folder.

“static” – all the css files and images reside in this folder.

“instance” – sqlite3 database reside in this folder.

“application” – this folder contains three files

1. “controllers.py” – contains all the application routers.
2. “database.py” – contains ‘db’ object.
3. “models.py” – contains all the database model classes.

Features implemented:

In this Library, a home page is created and further separate login forms for both librarian and general users are created. A register form is also created for new users.

Both librarian and user dashboards are created.

In the librarian dashboard, all CRUD operations can be implemented by librarian on both e-books and sections. Librarian can grant, reject, revoke the book access. He can track the user details of book request. Dynamic graphs are generated to improve the functionality.

In the user dashboard, user can view the available e-books, request, cancel the request, return, view the e-book, give the feedback when given access by librarian.

Search functionality is made available for both user and librarian.

Frontend validation is also implemented.

Bootstrap – used for all the forms and table generations and also small amount of css is used.

DB Schema Design:

Tables (5)

Name	Type	Schema
book		CREATE TABLE "book" ("id" INTEGER NOT NULL, "title" VARCHAR NOT NULL, "author" VARCHAR, "contents" TEXT, "section_title" TEXT, "section_id" INTEGER, PRIMARY KEY("id"), FOREIGN KEY("section_id") REFERENCES "section"("id"), UNIQUE("title"))
id	INTEGER	"id" INTEGER NOT NULL
title	VARCHAR	"title" VARCHAR NOT NULL
author	VARCHAR	"author" VARCHAR
content	TEXT	"content" TEXT
section_title	TEXT	"section_title" TEXT
section_id	INTEGER	"section_id" INTEGER
feedback		CREATE TABLE "feedback" ("id" INTEGER NOT NULL, "user_id" INTEGER, "book_id" INTEGER, "book_name" TEXT, "message" VARCHAR, FOREIGN KEY("book_id") REFERENCES "book"("id"), FOREIGN KEY("user_id") REFERENCES "user"("id"), PRIMARY KEY("id"))
id	INTEGER	"id" INTEGER NOT NULL
user_id	INTEGER	"user_id" INTEGER
book_id	INTEGER	"book_id" INTEGER
book_name	TEXT	"book_name" TEXT
message	VARCHAR	"message" VARCHAR
issue		CREATE TABLE "issue" ("id" INTEGER NOT NULL, "user_id" NUMERIC, "book_id" INTEGER, "book_name" TEXT, "section_name" TEXT, "status" VARCHAR NOT NULL, "request_type" VARCHAR, "requested_date" DATE NOT NULL, "return_date" DATE, "return_data" DATE, FOREIGN KEY("user_id") REFERENCES "user"("id"), FOREIGN KEY("book_id") REFERENCES "book"("id"), PRIMARY KEY("id"))
id	INTEGER	"id" INTEGER NOT NULL
user_id	NUMERIC	"user_id" NUMERIC
book_id	INTEGER	"book_id" INTEGER
book_name	TEXT	"book_name" TEXT
section_name	TEXT	"section_name" TEXT
status	VARCHAR	"status" VARCHAR NOT NULL
request_type	VARCHAR	"request_type" VARCHAR
requested_date	DATE	"requested_date" DATE NOT NULL
issue_date	DATETIME	"issue_date" DATETIME
return_date	DATE	"return_date" DATE
section		CREATE TABLE "section" ("id" INTEGER NOT NULL, "title" VARCHAR NOT NULL, "description" VARCHAR, "n_book" INTEGER DEFAULT 0, "create_date" DATE, UNIQUE("title"), PRIMARY KEY("id"))
id	INTEGER	"id" INTEGER NOT NULL
title	VARCHAR	"title" VARCHAR NOT NULL
description	VARCHAR	"description" VARCHAR
n_book	INTEGER	"n_book" INTEGER DEFAULT 0
create_date	DATE	"create_date" DATE
user		CREATE TABLE "user" ("id" INTEGER NOT NULL, "username" VARCHAR NOT NULL, "password" VARCHAR NOT NULL, "role" VARCHAR NOT NULL, PRIMARY KEY ("id"), UNIQUE ("username"))
id	INTEGER	"id" INTEGER NOT NULL
username	VARCHAR	"username" VARCHAR NOT NULL
password	VARCHAR	"password" VARCHAR NOT NULL
role	VARCHAR	"role" VARCHAR NOT NULL

The relational tables used for this project are:

section, user, book, issue, feedback.

One to many relationships are used between tables to implement the features for this project.

Foreign keys are specified to implement the one-to-many relationships between tables.

User can request, give feedback for multiple

books. A section can have multiple books.

A book can be issued multiple times.

Presentation video link: [Video](#)